

Feature Normalization and Regularization

COSHE Seminar

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Background

- Postdoctoral researcher at the University of Copenhagen since 2024
- PhD student at Lund University, graduated 2024
- Previous life as a physical therapist, including research (RCTs, meta-analyses, etc.)
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Interests

- High-dimensional statistics
- Regularization and sparsity
- Optimization and computational statistics
- Research software, reproducibility, data visualization, and data science in general

This Talk

I am going to present *The Choice of Normalization Influences Shrinkage in Regularized Regression* (Larsson and Wallin [2025](#)).¹

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- Feature normalization has large effects in regularized regression (lasso, ridge)
- But there is no research on this!

Conclusions

- Class balance has a normalization-dependent impact on regularized model.
- In mixed data, choice of normalization implicitly weighs features' importances.



Figure 1: Joint work with Jonas Wallin

¹Published in TMLR, to be presented at ICML 26.

Preliminaries and Motivation

A Little Theory

Experiments

Preliminaries and Motivation

- Data consists of a **fixed** matrix of features $\mathbf{X} \in \mathbb{R}^{n \times p}$ and a response vector $\mathbf{y} \in \mathbb{R}^n$.
- $\mathbf{y}$ comes from a linear model, that is,

$$y_i = \beta_0^* + \mathbf{x}_i^\top \boldsymbol{\beta}^* + \varepsilon_i \quad \text{for } i \in 1, \dots, n,$$

where $\boldsymbol{\beta}^*$ is the vector of *true* coefficients.

- ε_i is the measurement noise, generated from some random variable.

The Elastic Net

Linear regression plus a combination of the ℓ_1 (lasso) and ℓ_2 (ridge) penalties:

$$\beta^* = \arg \min_{\beta \in \mathbb{R}^p} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \underbrace{\lambda_1 \|\beta\|_1}_{\text{lasso}} + \underbrace{\frac{\lambda_2}{2} \|\beta\|_2^2}_{\text{ridge}} \right)$$

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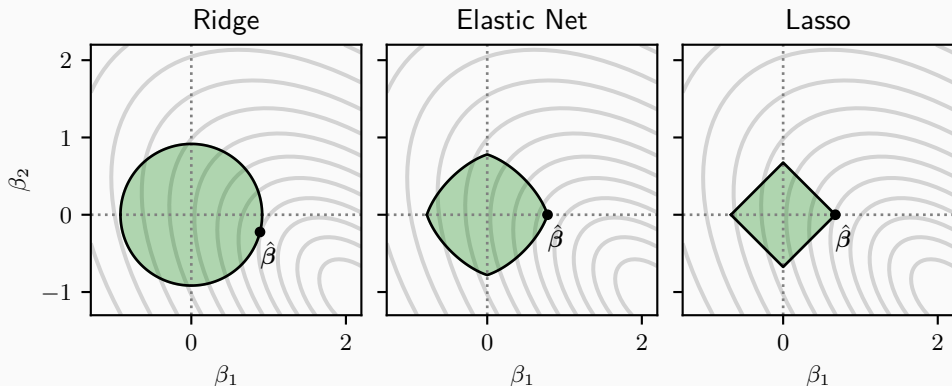


Figure 2: Elastic net is a combination of the lasso and ridge.

Regularization

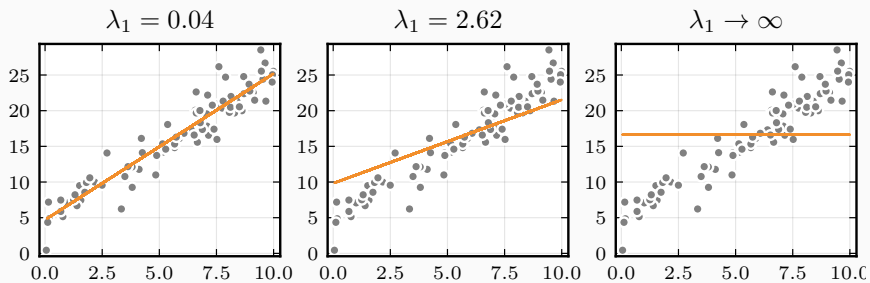


Figure 3: Regularization for a simple linear regression problem

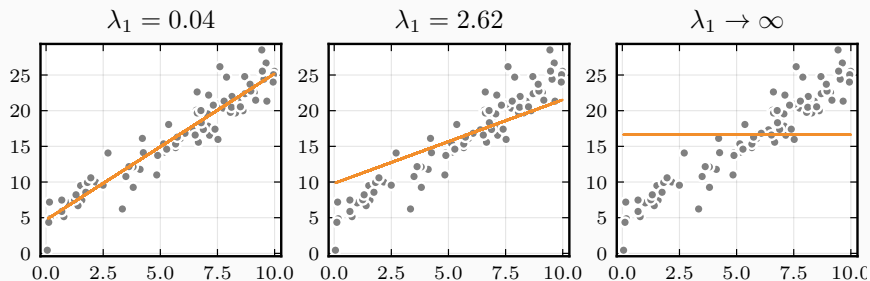


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Why Regularize? (Lasso, Ridge)

- Uniqueness when $p \gg n$
- To overcome overfitting

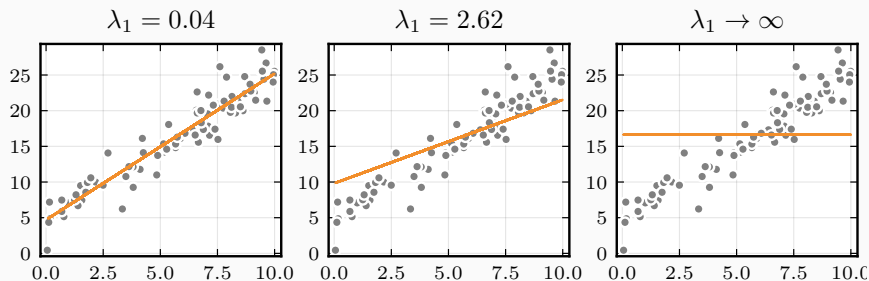


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Why Sparsity? (Lasso)

- Interpretability
- The sparsity bet

- Don't know optimal λ_1 (lasso) and λ_2 (ridge) in advance.

The Elastic Net Path

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with $\alpha \in [0, 1]$.

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$$\begin{aligned}\lambda_1 &= \alpha\lambda, \\ \lambda_2 &= (1 - \alpha)\lambda\end{aligned}$$

with $\alpha \in [0, 1]$.

- For each α , solve the elastic net over a sequence of λ : the **elastic net path**.

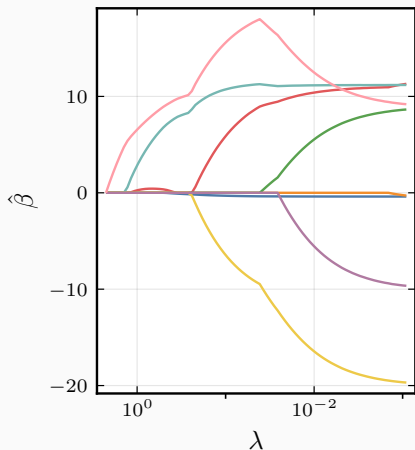


Figure 4: The elastic net path

Sensitivity to Scale

Lasso and ridge penalties are **norms**—feature scales matter!

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Example

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$$\mathbf{X} \sim \text{Normal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \right), \quad \boldsymbol{\beta}^* = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}.$$

and set $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}^* + \varepsilon$, with $\varepsilon \sim \text{Normal}(0, \sigma_\varepsilon^2)$.

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Large scale means less penalization because the size of β_j can be smaller for an equivalent effect (on $\mathbf{y}$).

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$$\tilde{x}_{ij} = \frac{x_{ij} - c_j}{s_j}.$$

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For interpretation, we typically transform the coefficients back to their original scale via

$$\hat{\beta}_j = \frac{\hat{\beta}_j^{(n)}}{s_j} \quad \text{for } j = 1, 2, \dots, p,$$

where $\hat{\beta}_j^{(n)}$ is a coefficient from the normalized problem.

Table 1: Common ways to normalize X

Normalization	Centering (c_j)	Scaling (s_j)
Standardization	$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$	$\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}$
Min–Max	$\min_i(x_{ij})$	$\max_i(x_{ij}) - \min_i(x_{ij})$
Max–Abs	0	$\max_i(x_{ij})$
Adaptive Lasso	0	β_j^{OLS}

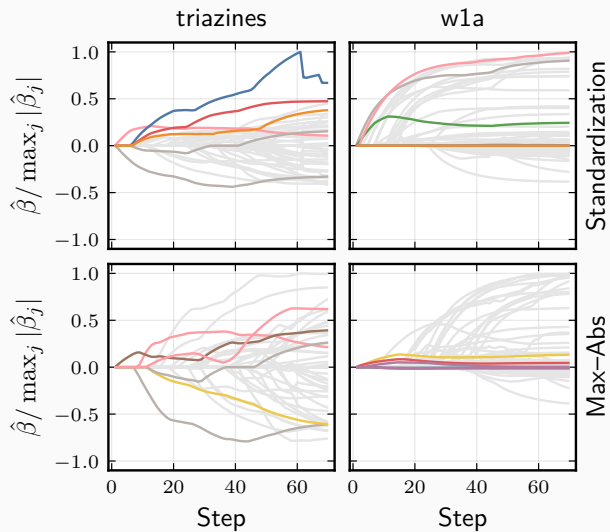


Figure 5: Normalization matters. Lasso paths under two different types of normalization (standardization and max-abs normalization). The union of the first ten features selected in any of the settings are colored.

Table 2: Lasso coefficients on test sets, with λ set from 5-fold cross-validation repeated 5 times. We show the five largest coefficients in magnitude for each data set for the standardization setting and corresponding coefficients for the max-abs normalization setting.

housing		triazines		w1a	
$\hat{\beta}_{\text{std}}$	$\hat{\beta}_{\text{max-abs}}$	$\hat{\beta}_{\text{std}}$	$\hat{\beta}_{\text{max-abs}}$	$\hat{\beta}_{\text{std}}$	$\hat{\beta}_{\text{max-abs}}$
-0.63	-0.68	0.17	0.0	1.8	0.0
-1.4	-0.78	0.069	0.0	1.8	0.78
0.27	0.0	0.028	0.0	1.8	0.63
-0.99	-0.34	0.071	0.0	1.4	0.080
2.8	3.1	0.029	0.0	1.7	0.0

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For the elastic net, you should scale the penalty weights—not the features.

When mixing binary and normal features, normalization implicitly weighs their importance.

A Little Theory

Orthogonal Features

There is no explicit solution to the elastic net problem in general.

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$$\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = \text{diag}(\tilde{\mathbf{x}}_1^\top \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_p^\top \tilde{\mathbf{x}}_p),$$

then there is:²:

$$\hat{\beta}_j = \frac{S_{\lambda_1}(\tilde{\mathbf{x}}_j^\top \mathbf{y})}{s_j(\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{x}}_j + \lambda_2)},$$

where

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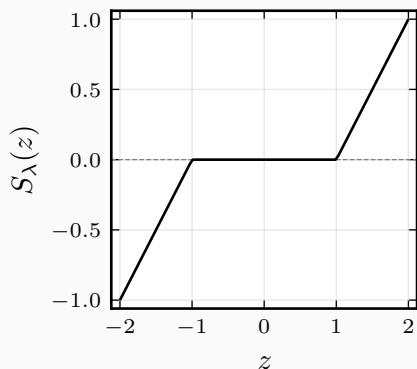


Figure 6: Soft thresholding

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Binary Features

Assume we have a binary feature x_j , such that $x_{ij} \in \{0, 1\}$. Let $q \in [0, 1]$ be the class balance of this feature, that is: $q = \bar{x}_j$.

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$s_j = q - q^2$ for lasso and $s_j = \sqrt{q - q^2}$ for ridge removes effect of q , which suggests the parametrization

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There is no (simple) s_j that will work for the elastic net.

Probability of Selection

Since $\mathbf{X}$ is fixed and ε is normal, we can compute the probability of selection.

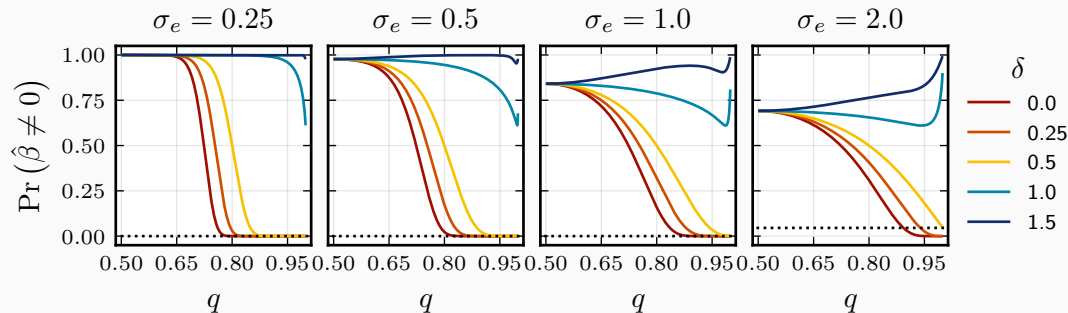
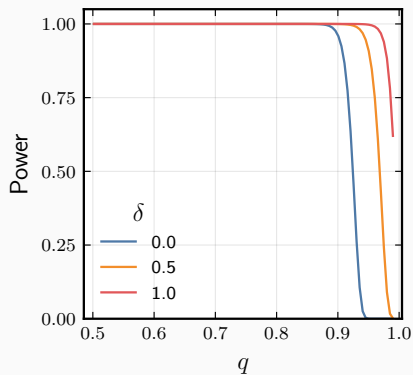
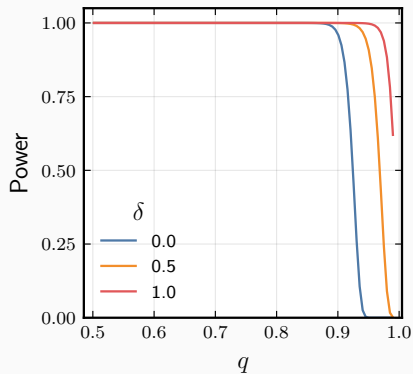


Figure 7: Probability that the elastic net selects a feature across different noise levels (σ_ε), types of normalization (δ), and class balance (q). The dashed line is asymptotic behavior for $\delta = 1/2$. Scaling used is $s_j \propto (q - q^2)^\delta$.

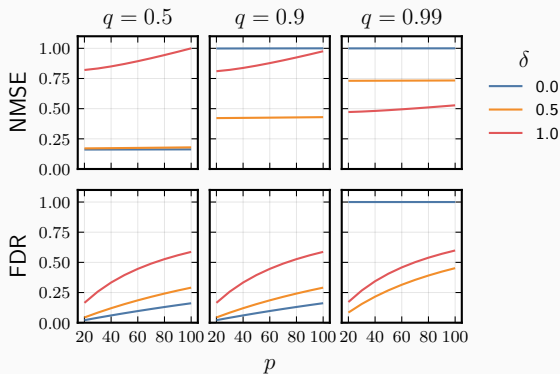


(a) Power in the sense of detecting all the true signals. Constant p .

Figure 8: Multiple features: 10 true signals and varying q and p . Mean squared error (MSE), false discovery rate (FDR), and power



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(b) False discovery rate (FDR) and normalized mean-squared error (NMSE).

Figure 8: Multiple features: 10 true signals and varying q and p . Mean squared error (MSE), false discovery rate (FDR), and power

Rare Traits

Features with large class-imbalances might not be selected even if effect is **very strong** (e.g. rare SNPs, mutations).

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Subgroup Data

Results become dependent on data collection.

Collecting more data with different class balances influences the results (since class balances change).

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How to put binary features and normal features on the “same” scale?

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How to put binary features and normal features on the “same” scale?

One Definition of Comparability

The effects of a binary feature and a normally distributed feature are **comparable** if a flip in the binary feature has the same effect as a two-standard deviation change in the normal feature (Gelman 2008).

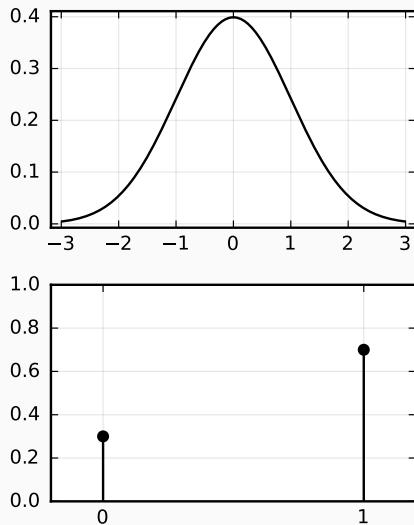


Figure 9: How do we match these?

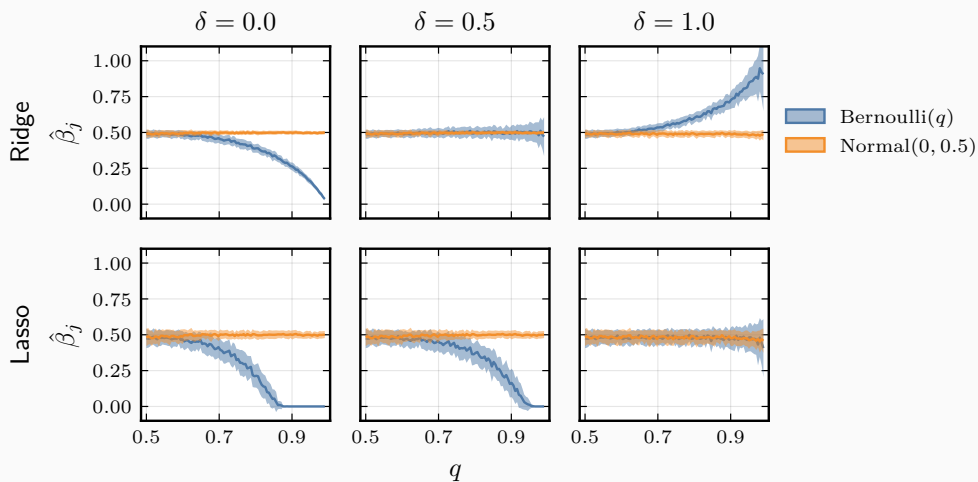


Figure 10: Comparison between lasso and ridge estimators for features generated to resemble features from various distributions.

Experiments

Binary Features (Decreasing q)

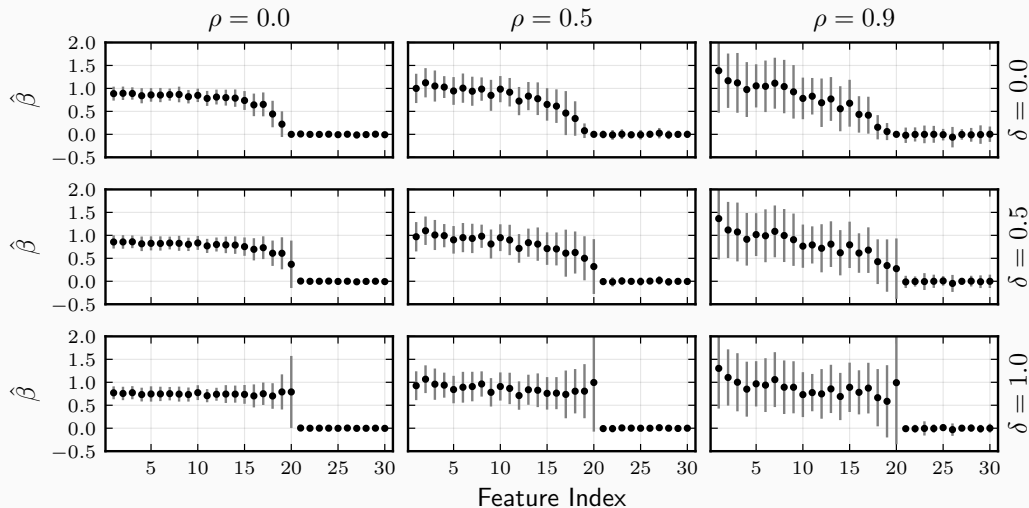


Figure 11: Lasso estimates for first 30 coefficients. First 20 features are true signals with a geometrically decreasing class balance from 0.5 to 0.99. ρ is a measure of autocorrelation.

Binary Features (Signal-to-Noise Ratio)

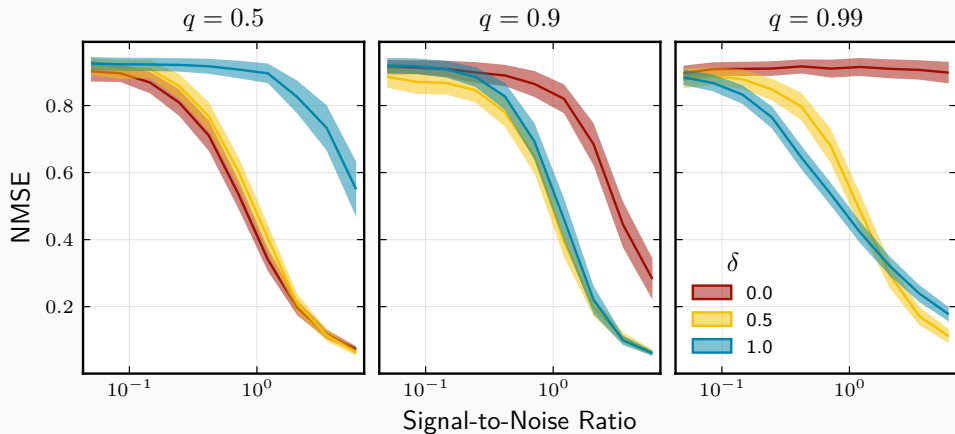


Figure 12: Normalized mean-squared test set error (NMSE).

Hyperparameter Optimization

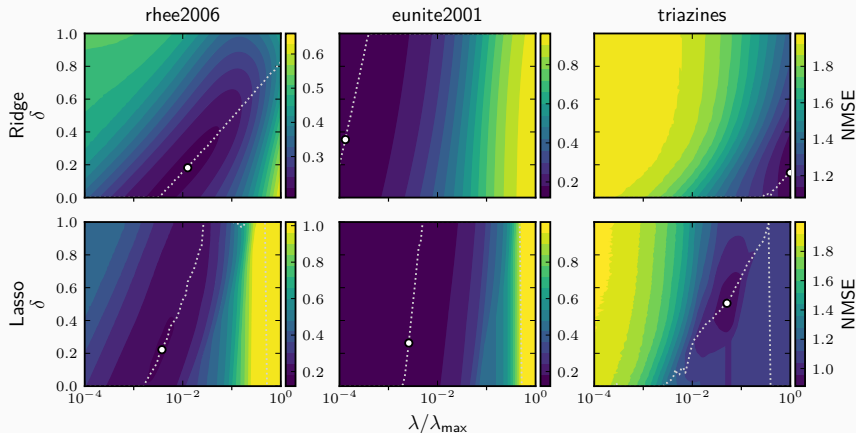


Figure 13: Contour plots of hold-out (validation set) error across a grid of δ and λ values for the lasso and ridge.

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Additional Findings

- Intersections of binary/normal features create additional issues
- The only way to control the bias–variance tradeoff in the elastic net is to scale the penalty weights, not the features.

Thank you!

-  El Ghaoui, Laurent, Vivian Viallon, and Tarek Rabbani (Sept. 21, 2010). *Safe Feature Elimination in Sparse Supervised Learning*. Technical report UCB/EECS-2010-126. Berkeley: EECS Department, University of California. URL: <http://www.eecs.berkeley.edu/Pubs/TechRpts/2010/EECS-2010-126.html>.
-  Gelman, Andrew (July 10, 2008). “Scaling Regression Inputs by Dividing by Two Standard Deviations”. In: *Statistics in Medicine* 27.15, pp. 2865–2873. ISSN: 02776715, 10970258. DOI: [10.1002/sim.3107](https://doi.org/10.1002/sim.3107). URL: <https://onlinelibrary.wiley.com/doi/10.1002/sim.3107> (visited on 09/27/2023).
-  Larsson, Johan and Jonas Wallin (Nov. 2025). “The Choice of Normalization Influences Shrinkage in Regularized Regression”. In: *Transactions on Machine Learning Research*. ISSN: 2835-8856. (Visited on 12/05/2025).

-  Tibshirani, Robert et al. (Mar. 2012). “Strong Rules for Discarding Predictors in Lasso-Type Problems”. In: *Journal of the Royal Statistical Society. Series B: Statistical Methodology* 74.2, pp. 245–266. ISSN: 1369-7412. DOI: [10/c4bb85](https://doi.org/10.1016/j.jrssb.2012.03.004). URL: <https://iths.pure.elsevier.com/en/publications/strong-rules-for-discarding-predictors-in-lasso-type-problems> (visited on 03/16/2018).
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Extras

Max-Abs Scaling of Continuous Features

- Min-max normalization is sometimes used in continuous data
- Very sensitive to outliers
- But also depend on sample size!
- In other words, results in model validation with varying sample sizes can yield very strange results.

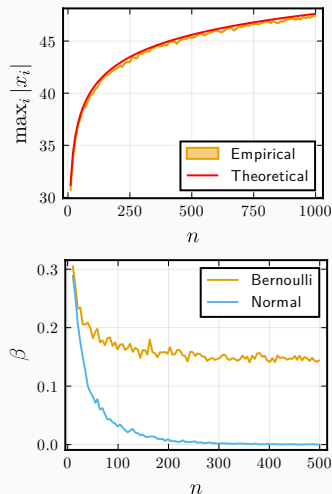


Figure 14: Effects of maximum absolute value scaling.

Hyperparameter Optimization (Support and NMSE)

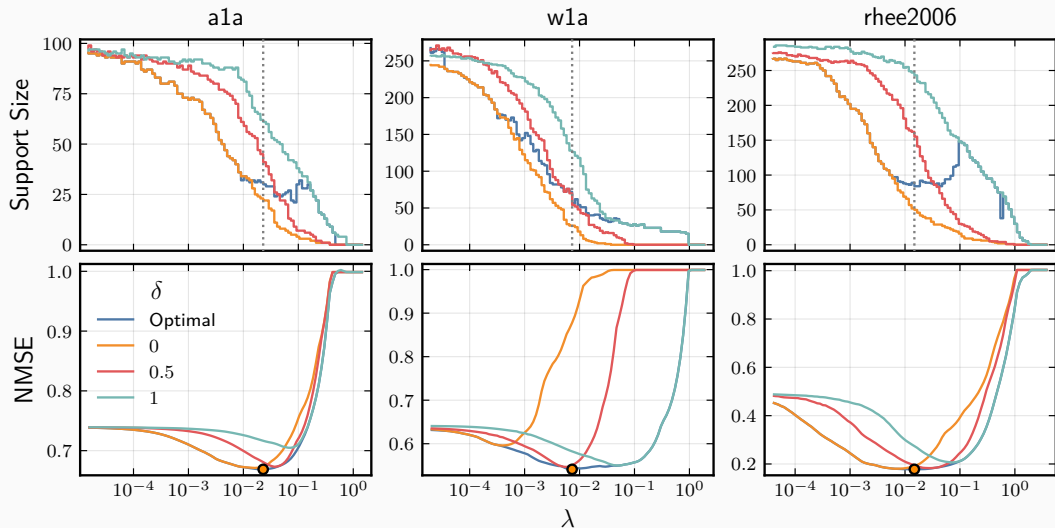


Figure 15: Support and NMSE of the lasso for different values of δ and λ .

- Proposed by Zou and Hastie ([2005](#)).

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- Lasso (ℓ_1) part:
 - Enables sparsity (interpretability, parsimony, feature selection)
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- Proposed by Zou and Hastie (2005).
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- Ridge (ℓ_2) part
 - Mitigates lasso issue in correlated data
 - Better predictive performance when true signal is non-sparse